
Where to Predict in Retinal OCT?

Anatomy-Guided Target Selection for I-JEPA

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Abstract

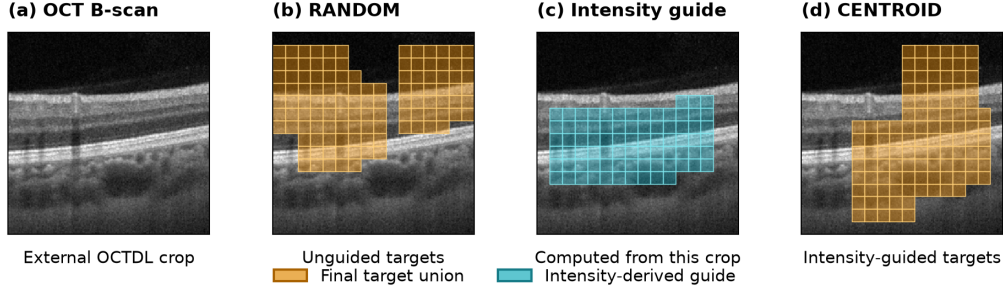
Learning transferable representations from retinal optical coherence tomography (OCT) is challenging because disease-related changes can be subtle and large, consistently labelled datasets require substantial expert annotation. Self-supervised learning offers a way to use unlabelled scans and reduce dependence on diagnostic labels. Image-based Joint-Embedding Predictive Architecture (I-JEPA) learns by predicting target-region representations from visible context rather than reconstructing pixels. However, its uniform target placement does not account for the layered organization of retinal OCT, where retinal tissue occupies only part of each B-scan. We investigate anatomy-guided target selection using image intensity and predicted retinal segmentation while retaining the I-JEPA architecture and latent prediction objective. We compare tissue-directed rectangular targets, anatomy-shaped targets and coverage-constrained placement with uniform masking, and evaluate the learned representations through frozen-encoder glaucoma classification on FairVision. Tissue-directed rectangle placement improves the observed classification results, including with segmentation-free intensity guidance, while more anatomically specific strategies do not consistently provide further gains. Analysis of the actual masks shows that target selection, target size and visible context are coupled. Our findings motivate anatomy-guided predictive learning for retinal OCT and better-controlled comparisons of what to predict and what to keep visible.

1 Introduction

Optical coherence tomography (OCT) reveals the layered structure of the retina and supports glaucoma assessment [Maetschke et al., 2019, Ran et al., 2019, Noury et al., 2022]. Self-supervised pretraining offers a way to learn from these images without requiring a diagnosis for every training scan [Azizi et al., 2021, Zhou et al., 2023, Qiu et al., 2024]. The spatial organization of OCT makes the choice of pretraining task particularly relevant: retinal tissue forms a relatively narrow region within a B-scan, surrounded by vitreous and other background.

Image-based Joint-Embedding Predictive Architecture (I-JEPA) [Assran et al., 2023] learns by predicting hidden image representations rather than reconstructing hidden pixels. A context encoder processes visible patches. A predictor uses these features and the target positions to estimate representations supplied by a target encoder. The target encoder is updated by an exponential moving average (EMA) of the context encoder. This creates a prediction task without diagnosis labels. Unlike masked autoencoders [He et al., 2022, Xie et al., 2022], I-JEPA does not require the predictor to reproduce each pixel.

Our interest in anatomical guidance also arose from an exploratory feature-occlusion analysis of glaucoma classifiers initialized from uniform-masking I-JEPA and subsequently fine-tuned with



OCT: Kulyabin et al., OCTDL Fig. 1 (CC BY 4.0), cropped. Illustration only; not FairVision.

Figure 1: Target-selection intuition on an external OCT example, not FairVision data. (a) One annotation-free crop from the real OCT B-scan in Kulyabin et al., OCTDL, Fig. 1 (arXiv:2312.08255v3), adapted under CC BY 4.0. (b) RANDOM final target union. (c) CENTROID’s intensity-derived guide. (d) CENTROID final target union. The same crop is downsampled to 256 by 256 pixels without contrast enhancement; masks use the production samplers, matched block-size draws and fixed seed. Overlays show exact delivered indices; encoder context is omitted. This single illustrative replay is not a segmentation annotation, historical training batch, evaluation sample or performance comparison. Source and license: <https://arxiv.org/abs/2312.08255v3> and <https://creativecommons.org/licenses/by/4.0/>.

diagnosis labels. Selected attribution maps showed prediction sensitivity near retinal structures: removing patch features from the pooled representation changed the classifier logit. This observation motivated asking whether retinal anatomy could guide target selection during pretraining (Appendix D).

Standard I-JEPA places rectangular targets uniformly across the image. In retinal OCT, many such targets include little retinal tissue. This raises our central question: *can I-JEPA benefit from selecting prediction targets according to retinal anatomy rather than uniformly across the image?* Figure 1 illustrates the change in target placement. Semantic and anatomy-guided masked modelling provide positive motivation for this question [Li et al., 2022, Kakogeorgiou et al., 2022, Li et al., 2024, Wang et al., 2025]. However, selecting a target also removes information from the context. The useful learning task depends on both what is predicted and what remains visible.

We investigate this question through five masking families, including segmentation-free intensity guidance and retinal segmentation from MIRAGE [Morano et al., 2025]. These strategies vary target location, shape, or anatomical coverage while retaining the I-JEPA architecture and latent prediction objective. We evaluate the resulting B-scan representations by pooling them across an OCT volume and fitting a linear glaucoma classifier. The study includes six implementations, with two versions of anatomy-shaped masking.

Tissue-directed rectangle placement gives higher AUC than uniform masking in the reported runs. More detailed anatomical targeting does not consistently improve on these simpler strategies. To understand what the comparisons test, we measure the target and context patches that actually reach the model. This analysis identifies differences in target size, retained context and repeated target indices, as well as a coverage-scoring mismatch. We therefore present comparisons of implemented masking strategies, not controlled ablations of anatomical coverage.

Contributions.

- An empirical comparison of intensity-guided, segmentation-guided and uniform target selection for I-JEPA on retinal OCT.
- Frozen-probe results showing gains from tissue-directed rectangle placement, alongside mixed results for more specific anatomical targeting.
- An analysis of the actual target and context masks that identifies implementation constraints for better-controlled guided-masking studies.

67 2 Related work

68 **Masked image modelling and JEPAs.** Masked autoencoders reconstruct pixels [He et al., 2022,
69 Xie et al., 2022], whereas BEiT predicts discrete visual tokens [Bao et al., 2022] and iBOT distills
70 teacher outputs at masked patch positions [Zhou et al., 2022]. Data2vec and I-JEPA learn by pre-
71 dicting latent representations [Baevski et al., 2022, Assran et al., 2023]. Related approaches extend
72 masked prediction to video and volumetric data [Bardes et al., 2024, Chen et al., 2023b, Taleb et al.,
73 2020]. JEPA formulations also differ in their objectives and prediction structure [Balestriero and
74 LeCun, 2025, He et al., 2025]; we retain the I-JEPA objective and study target selection.

75 **Region selection and prediction structure.** DSeq-JEPA combines attention-derived region se-
76 lection with sequential prediction [He et al., 2025]. Its highest-ranked region supplies context for
77 predicting subsequent regions. Its component ablation favors the joint design rather than region
78 selection alone. Our study instead uses retinal anatomy and parallel I-JEPA prediction. This dis-
79 tinction matters: a guidance map, the selected targets and the available context jointly define the
80 learning task. I-JEPA’s own masking analysis emphasizes both sufficiently large targets and infor-
81 mative context [Assran et al., 2023].

82 **Medical region-conditioned prediction.** US-JEPA uses a frozen ultrasound teacher, and its
83 region-conditioned variant restricts both context and target sampling to valid ultrasound con-
84 tent [Radhachandran et al., 2026]. Our study addresses a different question: how intensity-based
85 and retinal-segmentation guidance change target selection within the EMA-based I-JEPA framework
86 for OCT. We compare simple rectangle placement with anatomy-shaped and coverage-constrained
87 targets, without introducing sequential prediction or a new teacher architecture. The contribution is
88 this OCT comparison and analysis of the resulting prediction tasks, not the general idea of informed
89 masking.

90 **Informed masking.** ADIOS learns adversarial masks [Shi et al., 2022]; SemMAE uses semantic
91 parts [Li et al., 2022]; AttMask selects high-attention patches [Kakogeorgiou et al., 2022]; and
92 HardPatches and AutoMAE adapt masking to prediction difficulty [Wang et al., 2023, Chen et al.,
93 2023a]. Medical approaches include AnatoMask, AnatomyMAE, MSMAE and VasoMIM [Li et al.,
94 2024, Ceballos Arroyo et al., 2026, Mao et al., 2023, Huang et al., 2026]. These studies motivate
95 informed masking, but their objectives, guidance sources and context construction differ. SemMAE,
96 for example, gradually changes from within-part to whole-part prediction rather than masking whole
97 parts from the start. We study anatomy-guided target selection under latent prediction on retinal
98 OCT.

99 **Retinal foundation models and OCT.** RETFound [Zhou et al., 2023], VisionFM [Qiu et al.,
100 2024], URFound [Yu et al., 2024], and OCTCube [Liu et al., 2026] pretrain on large retinal cor-
101 pora. OCT-specific masked modelling is explored by Pissas et al. [2024]. MIRAGE [Morano et al.,
102 2025] provides multimodal OCT segmentation and supplies the anatomical guidance used by sev-
103 eral of our arms, and deep glaucoma detection from OCT is established [Maetschke et al., 2019, Ran
104 et al., 2019, Noury et al., 2022]. FairVision provides the glaucoma-labelled OCT data used for our
105 downstream comparison [Luo et al., 2024].

106 3 Background and target-selection framework

107 We first define the I-JEPA prediction task, then describe how each masking strategy selects its targets.
108 The architecture and latent prediction loss are shared; differences in guide sources, target size and
109 final context are part of the strategy comparisons.

110 3.1 I-JEPA preliminaries

111 Let x be a B-scan partitioned into N patches, and let C denote the indices of the patches supplied to
112 the context encoder. Target group T_m contains K_m patch indices, with $m = 1, \dots, M$. We treat T_m
113 as an index list because some implementations repeat indices when equalizing target lengths. The
114 context excludes the union of target positions. Writing p_C and p_{T_m} for the corresponding positional

115 embeddings, the context and prediction paths are

$$h_C = f_\theta(x; C), \quad \hat{z}_m = g_\phi(h_C, p_C, p_{T_m}). \quad (1)$$

116 Here $f_\theta(x; C)$ embeds only the selected patches at their original image positions. The predictor
 117 projects h_C to its own feature dimension and adds its own positional embeddings p_C ; target queries
 118 combine a learned mask token with p_{T_m} . The target encoder sees the full B-scan. Its output is
 119 normalized per patch before selecting the target positions:

$$z_m = \text{sg}([\text{LN}(f_{\bar{\theta}}(x))]_{T_m}), \quad \bar{\theta} \leftarrow \mu \bar{\theta} + (1 - \mu)\theta. \quad (2)$$

120 The operator sg stops gradients, LN is LayerNorm across feature dimensions, and μ is the EMA co-
 121 efficient. The predictor uses the same context for each target group; predictions are not conditioned
 122 on earlier target predictions.

123 Each target and prediction is a $K_m \times D$ matrix. For feature dimension D , the implementation
 124 minimizes the mean Smooth-L1 loss over target instances and feature coordinates:

$$\mathcal{L} = \frac{1}{D \sum_{m=1}^M K_m} \sum_{m=1}^M \sum_{k=1}^{K_m} \sum_{d=1}^D \rho(\hat{z}_{m,k,d} - z_{m,k,d}). \quad (3)$$

125 The Smooth-L1 function ρ uses the standard unit transition between its quadratic and linear regions.
 126 Repeated indices contribute repeated loss terms. The code equalizes target lengths within a batch
 127 and applies a single mean reduction to the stacked prediction and target tensors, including averaging
 128 over images. Figure 2 shows the pretraining and evaluation paths.

129 3.2 Anatomy-guided target selection

130 For strategy s , let $A_s(x)$ denote its patch-grid guidance: an intensity-derived band for CENTROID,
 131 a repaired segmentation envelope for ENVELOPE, or soft retinal-class score maps for the anatomy-
 132 shaped and coverage strategies. These signals need not have the same channel structure. CENTROID
 133 is an intensity-guided tissue-localization baseline, not a segmentation model. We write the joint
 134 masking procedure as

$$(C, T) \sim \Pi_s(\cdot \mid A_s(x)), \quad T = (T_1, \dots, T_M). \quad (4)$$

135 The random baseline uses geometric block sampling without consulting image guidance. Π_s in-
 136 cludes target placement and context construction, including batch-dependent token equalization. A
 137 candidate context block is sampled, target positions are excluded, and token lists are shortened as
 138 needed for batching. Only the remaining context tokens enter f_θ . Consequently, tissue outside the
 139 target union need not be visible to the encoder. We define *mask ratio* as the number of unique target
 140 positions divided by N , and *tissue coverage* as the fraction of guide-positive patches included in that
 141 union. Neither is the fraction of the image supplied as context.

142 3.3 Compared masking strategies

143 All policies request $M=4$ target blocks and one context block. Rectangle policies share nominal
 144 scale and aspect-ratio ranges, but placement and collation can change unique target area and encoder-
 145 visible context.

146 **RANDOM (baseline).** Stock I-JEPA multiblock. Four rectangles placed uniformly at random any-
 147 where in the grid. No image content is consulted.

148 **ENVELOPE (random-within-retina).** The same nominal rectangle count, scale and aspect-ratio dis-
 149 tributions as RANDOM, but rejection-sampled onto the MIRAGE-derived repaired retinal
 150 envelope [Morano et al., 2025]. The predicted inner-retina and choroid union is repaired
 151 across the unlabeled middle-retina gap in native label space, then transformed jointly with
 152 the image. This changes the placement policy, not target shape. Overlap and delivered
 153 context are consequences of that intervention, not independently matched controls.

154 **CENTROID (segmentation-free).** A band whose vertical position is located per slice by a per-
 155 column intensity-weighted row centroid, smoothed across columns, with no segmentation
 156 model and no ground-truth annotation of any kind. This band guides the placement of four
 157 rectangular targets; it is not itself the target shape.

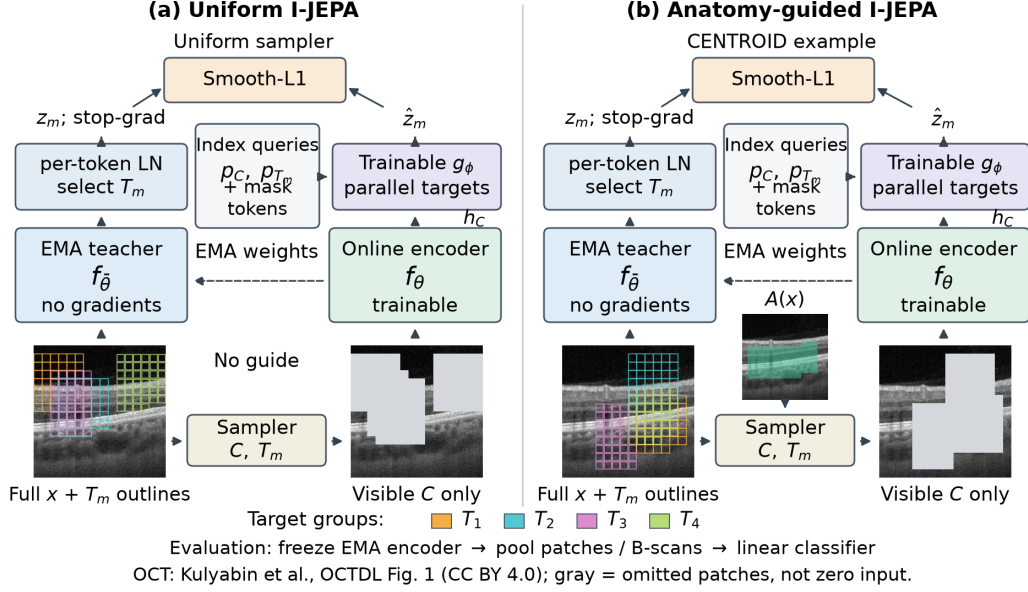


Figure 2: Uniform and anatomy-guided target selection within the same I-JEPA framework. In both panels, the context encoder and predictor are trainable. The EMA target encoder receives the full image; normalized target embeddings are selected afterward with gradients stopped. The predictor uses visible context and positional queries to predict each target group in parallel, with the same Smooth-L1 objective. The panels differ in target selection and the resulting visible context, not the architecture or loss. The guided panel illustrates CENTROID, not a relevance-ranking or sequential-prediction method. Masked views visualize omitted tokens rather than zero-filled encoder inputs. The image and exact masks reuse the licensed example in Figure 1; colours identify target groups, not relevance. The shared footer shows frozen-encoder evaluation.

- 158 **ANATOMY-V1 / ANATOMY-V2.** Ragged connected components grown on the MIRAGE score map,
 159 so target *shape* follows tissue rather than being rectangular. v2 also bridges diagonal adja-
 160 cencies.
- 161 **COVER f .** Targets chosen greedily to cover anatomy while leaving a fraction f of supported soft
 162 guide mass outside the target union. The historical run uses $f=0.21$. This condition
 163 does not guarantee the same fraction survives final encoder-context collation, and post-
 164 placement target truncation changes the intended coverage (Appendix B).

165 4 Experimental setup

166 **Data.** We use FairVision OCT volumes with binary glaucoma labels [Luo et al., 2024] and the
 167 fixed released splits. Training and validation contain 6,000 and 1,000 volumes, respectively. Each
 168 volume contributes 100 B-scans, resized to 256×256 . The test split contains $N=3000$ volumes
 169 (1466 positive). Diagnosis labels are used only for downstream evaluation, not JEPA pretraining.
 170 Demographic attributes are not model inputs.

171 **Pretraining.** Patch-level I-JEPA, ViT-Base, 256×256 crops, patch size 16, predictor depth 6,
 172 EMA target encoder. The reference fork is the epoch-25 checkpoint, with matched nominal opti-
 173 miser, learning-rate and weight-decay schedules and effective batch size (512). Guidance is ramped
 174 linearly from epoch 25 to 30 and held at full strength thereafter. The early ANATOMY-V1 result has
 175 incomplete launch documentation; its configuration resumes an ENVELOPE epoch-27 checkpoint
 176 rather than directly documenting an epoch-25 fork. The encoder processes B-scans independently,
 177 without learned cross-slice attention. Rectangle strategies share nominal target scales, but the guide
 178 products and realized masks are not identical. ANATOMY-V2 also uses fixed-length targets. These
 179 differences are documented in Appendix A.

Table 1: Frozen MeanPool probe, FairVision test ($N=3000$, 1466 positive). The reference ancestor is epoch 25 (AUC 0.8487); continuation provenance is described in Section 4. Δ is versus RANDOM at the *matched* epoch. The upper block is precision-matched (fp16 throughout) and carries every headline claim; the lower block is fp32, each Δ taken against an fp32 null re-probed at the same epoch. The measured fp16/fp32 gap is at most 2×10^{-4} (Appendix F).

policy	guidance signal	epoch 50		epoch 75		epoch 100	
		AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	none (null)	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	segmenter (location only)	0.8761	+0.0120	0.8803	+0.0080	0.8807	+0.0062
CENTROID	intensity centroid, no segmenter	0.8740	+0.0099	0.8836	+0.0113	0.8855	+0.0109
ANATOMY-V2	segmenter (target shape)	0.8654	+0.0013	0.8612	−0.0111	—	—
COVER $f=.21$	segmenter (coverage)	0.8643	+0.0002	0.8639	−0.0084	0.8577	−0.0168

180 RANDOM, CENTROID, ENVELOPE and COVER reach epoch 100. ANATOMY-V2 reaches epoch 75;
 181 its epoch-100 result is unmeasured. ANATOMY-V1 has only an epoch-30 result. We compare strate-
 182 gies at matched pretraining epochs, with epoch 50 as the common endpoint for the five main policies.
 183 Excluded continuations are listed separately in Appendix A.

184 **Evaluation.** The EMA target encoder is frozen. We average patch features within each B-scan,
 185 then average across the volume and train a LayerNorm–linear head. Each head uses a 50-epoch
 186 budget and is selected by validation AUC. We report test AUC with percentile bootstrap intervals
 187 and paired differences (10,000 class-stratified resamples, fitted heads fixed), together with DeLong
 188 tests [DeLong et al., 1988]. Paired analyses rely on the documented shared split and determinis-
 189 tic loader ordering; historical prediction files lack independent case-ID manifests. These intervals
 190 describe test-case sampling, not variation across pretraining runs.

191 **Numerical precision.** The original rectangle-policy probes use mixed precision, while the
 192 anatomy-shaped and coverage-policy probes use fp32. All primary differences in Table 1 use a
 193 baseline matched in both epoch and probe precision. Re-probes quantify the small numerical differ-
 194 ences (Appendix F); probe precision is distinct from pretraining target-encoder precision.

195 5 Results and analysis

196 We first compare downstream classification, then examine the actual masks to determine which
 197 properties changed. These are strategy comparisons, not factorial ablations: each policy has one
 198 pretraining continuation, and some policies change several aspects of the prediction task.

199 5.1 Comparison of target-selection strategies

200 **Tissue-directed rectangles.** ENVELOPE exceeds uniform masking by +0.0120 AUC at epoch 50.
 201 The segmentation-free CENTROID strategy exceeds it by +0.0109 at epoch 100. Both strategies have
 202 positive paired differences from RANDOM at all three evaluated epochs (Figure 3). The full intervals
 203 and multiple-comparison adjustment are reported in Appendix E. These gains support tissue-directed
 204 target placement in the observed runs. They do not require targets to contain only retinal tissue: the
 205 large rectangles also include background (Table 2).

206 **Anatomy-shaped targets.** At the matched epoch 50, ANATOMY-V2 has lower AUC than ENVE-
 207 LOPE ($\Delta = -0.0107$; CI $[-0.0167, -0.0046]$). At epoch 30, however, ANATOMY-V1 sits +0.0044
 208 *above* it (CI $[+0.0009, +0.0078]$), with the early continuation’s lineage uncertainty described in
 209 Section 4. At epoch 50, the intervals for both ANATOMY-V2 and COVER versus RANDOM include
 210 zero.

211 **Coverage-constrained placement.** COVER improves to epoch 50, peaks at epoch 73, then *falls*,
 212 losing -0.0071 from its own peak ($p < 0.0001$) while the unguided null keeps improving (Figure 3).
 213 Although COVER selects rectangles by anatomical coverage, batching shortens the targets after place-

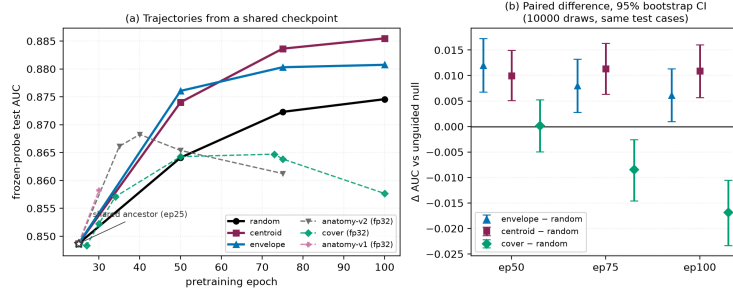


Figure 3: **(a)** Frozen-probe test AUC against pretraining epoch; note the truncated AUC axis. The open star marks the shared epoch-25 ancestor. Solid lines are the precision-matched fp16 family; dashed lines are fp32 arms and must not be compared vertically against the solid lines. **(b)** The actual inference: paired differences against the unguided null on identical test cases, 10,000-resample percentile bootstrap intervals. Every ENVELOPE and CENTROID interval excludes zero; COVER straddles zero at epoch 50 and is negative thereafter. Paired differences account for the covariance of predictions on shared test cases; individual-arm intervals answer a different question.

Table 2: Per-image mask geometry on 600 training slices using the production samplers, before batch-level context truncation. Tissue statistics use a thresholded MIRAGE-derived proxy, not ground-truth retinal annotations. “purity” is the fraction of masked cells lying on tissue; “loss slots” counts supervised target-token instances, summed over the M targets and including duplicates. AUC is quoted at the *matched* epoch 50, so the geometry-to-AUC association is not read across mixed epochs. Final encoder-context fractions at the production batch size are reported in Table 3.

policy	anatomy hidden	purity	mask ratio	context kept	loss slots	AUC @ep50
RANDOM	54.0%	31.5%	44.5%	41.9%	159.9	0.8641
CENTROID	62.1%	40.0%	40.3%	45.5%	159.0	0.8740
ENVELOPE	77.6%	43.3%	46.5%	40.6%	159.7	0.8761
COVER	73.5%	44.2%	43.2%	43.2%	159.1	0.8643
ANATOMY-V2	79.9%	97.1%	21.3%	67.7%	64.0	0.8654

ment. Its final masks therefore hide less anatomy than ENVELOPE’s (Table 2), rather than providing a higher-coverage comparison. Appendix B describes this target-sizing behavior.

CENTROID has the highest epoch-100 point estimate, exceeding ENVELOPE by +0.0047 (CI [+0.0004, +0.0091]). The paired CENTROID–ENVELOPE difference intervals include zero at epochs 50 and 75. Thus a simple intensity-based strategy is a useful comparator for segmentation-guided methods, not an anatomical oracle.

5.2 Analysis of target and context masks

The samplers do not differ only in how much tissue they target. ANATOMY-V2 has high tissue purity but a smaller target union and more total context than the rectangle strategies (Table 2). Its fixed $K=16$ indices per target give $4 \times 16 = 64$ prediction instances, compared with about 159 for the rectangle strategies. This is an explicit target-length choice, not an inherent property of irregular shapes. Repeated target indices also affect loss weighting. The anatomy-shaped and coverage strategies use different guide products from the historical ENVELOPE run (Appendix A).

We therefore report target and context geometry alongside AUC. Appendix A gives the per-image measurement protocol; Appendix C measures the final masks in complete training-sized batches. Together, these distinguish tissue outside the target union from tissue actually supplied to the context encoder.

6 Discussion and limitations

The positive rectangle-policy results motivate anatomy-guided target selection without requiring a new encoder architecture. The segmentation-free centroid strategy is particularly useful as a simple baseline. More specific strategies also changed target size, guide source and retained context. Future comparisons should control these factors while varying the target-selection rule.

Experimental limits. Each policy has one pretraining continuation, so test-case confidence intervals do not establish a ranking over retrainings [Bouthillier et al., 2021]. The early anatomy-shaped result has incomplete continuation documentation, and ANATOMY-V2 was stopped after inspecting its epoch-75 deficit. Its later trajectory is unknown. Probe optimization and initialization can also affect frozen-probe comparisons; the reported fixed-budget protocol is not a measurement of the best achievable classifier for each encoder. Repeated pretraining comparisons remain future work.

The study uses one dataset and one downstream task. Policies, checkpoints and analyses were chosen after repeated inspection of the same test split, although probe heads were fitted on training data and selected on validation data. Reported intervals are therefore descriptive rather than confirmatory. Segmentation accuracy on the OCT pretraining data is not independently established. Only the specified coverage setting was pretrained, so we identify neither an optimal coverage level nor a mechanism for its AUC decline.

Broader impact and ethics. This study uses a public, de-identified retinal dataset under its release terms and involves no new recruitment or data collection, and no model reported here is clinically validated or intended for deployment (Appendix A).

7 Conclusion

We investigated where I-JEPA should predict in retinal OCT. Tissue-directed rectangle placement improves frozen-probe glaucoma AUC in the observed runs, including a simple intensity-centroid strategy. More anatomically specific targeting does not consistently improve these results, and the implementations differ in more than target location. The next step is to compare target selection under matched guidance, target sizes and visible context across repeated pretraining runs.

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397 A Additional experimental details

398 **Pretraining and linear evaluation.** The input crop is 256×256 , with patch size 16 and a ViT-Base
399 encoder. The predictor has depth 6 and hidden dimension 384. Pretraining uses AdamW with peak
400 learning rate 0.00025, a five-epoch warmup, final learning rate 0.000001, weight decay scheduled
401 from 0.04 to 0.4, and EMA coefficient from 0.996 to 1.0. The effective batch size is 512, formed
402 from batches of 64 with gradient accumulation. Rectangle target scales are 0.15–0.20 of image
403 area, aspect ratios 0.75–1.5, and context scales 0.85–1.0. Random resized crops use area scale
404 0.3–1.0. The guidance probability increases over the epoch-25 to epoch-30 continuation window;
405 fixed-length anatomy targets use $K = 16$ throughout. The frozen linear head uses AdamW, learning
406 rate 0.0004, weight decay 0.05, batch size 256, seed 42, and a 50-epoch budget with validation-AUC
407 selection and patience 15.

408 **Guide sources.** ENVELOPE uses the MIRAGE-derived repaired hard retinal envelope. The
409 anatomy-shaped and coverage strategies use precomputed soft guides from MIRAGE with frozen
410 residual adapters. The anatomy-v1 guide uses a decoder-feature adapter; anatomy-v2 and COVER
411 use an encoder-feature adapter. These guide products are not independently controlled in the cross-
412 strategy comparison. CENTROID uses image intensity rather than a segmentation model. The
413 adapters optimize feature-similarity objectives on FairVision-derived image caches, not a glaucoma
414 classification loss, and are frozen during I-JEPA training. The retained provenance does not inde-
415 pendently establish subject-level non-overlap for all data used to train the guidance models.

416 **Included and excluded continuations.** The early anatomy-v1 configuration resumes an ENVE-
417 LOPE epoch-27 checkpoint; its producing launch record is incomplete. Anatomy-v2 results use
418 the continuation with full-precision target-encoder outputs. Probes from its earlier mixed-target-
419 precision continuation are excluded, as are earlier uniform controls that used different target preci-
420 sion and context selection. The main table retains the comparable policy endpoints, including the
421 adverse COVER trajectory.

422 **Mask measurement.** Table 2 uses the seed-42 measurement on 600 training slices from 24 vol-
423 umes, with 25 slices per volume. The samplers are evaluated one image at a time on a 16×16 patch
424 grid, using guide occupancy at threshold 0.25. The “context kept” column is the per-image context;
425 batching can shorten it further. Table 3 records that difference.

Table 3: Visible context as a fraction of the patch grid, same 600 slices and same seed, measured per image and as delivered to the encoder at the production batch size of 64. The per-image column reproduces Table 2’s “context kept” to within 0.4 points.

policy	per image	delivered at batch 64
RANDOM	41.9%	24.7%
CENTROID	45.5%	32.9%
ENVELOPE	40.6%	30.7%
COVER	43.2%	30.1%
ANATOMY-V2	67.7%	63.5%

Data use and ethics. This is secondary research on the public, de-identified FairVision release under its CC BY-NC-ND 4.0 data licence. No new participants were recruited or identifying fields added. A separate institutional ethics or exemption determination is not recorded for this project. The models are not clinically validated or intended for patient care. The illustrative OCT image in the main paper comes from the separately attributed CC BY 4.0 OCTDL publication, not from the FairVision evaluation cohort.

B Target sizing and visible context

Batch equalization. Predictor targets are shortened to the minimum target length in the micro-batch. COVER chooses placements using the anatomical mass inside full candidate rectangles before this equalization. Because target indices are ordered row-major, shortening removes the lower portion of each rectangle rather than a spatially random subset. Coverage at placement time can consequently differ from coverage at the loss.

Measured geometry. The retained floor sweep over 6,137 slices gives anatomical coverage of 73.1% for COVER and 77.6% for ENVELOPE. Across 24,000 emitted targets, 73.4% remain perfect rectangles. Thus the trained COVER policy is not a higher-coverage counterpart to ENVELOPE. Table 2 reports the final target sets, and Appendix C provides a fixed-crop batch replay.

ENVELOPE shares the same shortening rule but does not optimize an exact coverage level. The anatomy-shaped strategies instead resample targets to a fixed $K=16$ indices. Equalization therefore has different consequences across the strategies, even when nominal target counts agree.

C Final target and context masks

Scope. We replayed the actual guided dataset, paired transforms, DataLoader worker collator and returned mask tensors on the existing 600-view Training scope (24 volumes, 25 views each), using fixed new crop draws shared across policies. These are not reconstructed historical training crops. Drawn block sizes were injected independently of placement; equal seeds alone do not pair policy-dependent rejection sampling. The production-size subset contains 576 views in complete batches of 64, with 573 valid post-crop guides. The shorter final batch is excluded from Table 4. Views within volumes are correlated, so these are descriptive mask statistics, not independent training replications.

What is measured. Target coverage is measured both at scoring and after target collation. Encoder-visible tissue is counted only in the final encoder indices, not in the full-grid target complement. Tissue is a guide occupancy thresholded at 0.25, not clinical ground truth. The context criterion is at least $\max(\lceil 0.21N_{\text{tissue}} \rceil, \min(4, N_{\text{tissue}}))$ non-target tissue cells. This occupancy-cell diagnostic is distinct from the policy’s soft-mass constraint on the target complement.

Random fill is not a background quota. At full ramp all usable targets take the guided branch, while the `random_legal` fill mechanism is conditional on unused target blocks. Some views have no random-fill blocks. Such blocks can include tissue, and guided blocks can include background, so branch labels cannot stand in for background supervision. Target composition is therefore measured from the returned token indices rather than inferred from the branch name.

Table 4: Final masks from the trained COVER strategy on 576 fixed training views. Context-criterion misses count only the 573 valid guides. Scored and final coverage refer to supported soft guide mass; context counts include all views.

Measurement	COVER
Views losing tissue targets during equalization	300
Valid-guide context-criterion misses	358
Views with zero tissue context	55
Scored hidden guide mass (%)	78.51
Final hidden guide mass (%)	74.57
Mean encoder-visible tissue cells	10.13

463 D Motivating attribution analysis and evidence limits

464 **Origin of the target-selection question.** Before studying anatomy-guided masking, we inspected
 465 feature-occlusion maps from glaucoma classifiers initialized with the uniformly masked I-JEPA
 466 baseline and then fine-tuned. The analysis compared mean pooling, cross-attention pooling and
 467 an attentive probe. At the B-scan level, a slice representation was zeroed before re-evaluating the
 468 classifier. At the patch level, an encoded patch was omitted from its B-scan’s patch mean, the mean
 469 was recomputed from the remaining patches, and the altered slice representation was passed to the
 470 pooling module and classification head. The change in logit measures sensitivity to this feature
 471 intervention; it is not pixel occlusion or an attention map.

472 Let F denote the sequence of B-scan representations supplied to the classifier and $\ell(F)$ its glaucoma
 473 logit. If $F^{(-s,p)}$ replaces the representation of B-scan s with the mean of its remaining patch features
 474 after omitting patch p , the attribution is

$$\Delta_{s,p} = \ell(F) - \ell(F^{(-s,p)}). \quad (5)$$

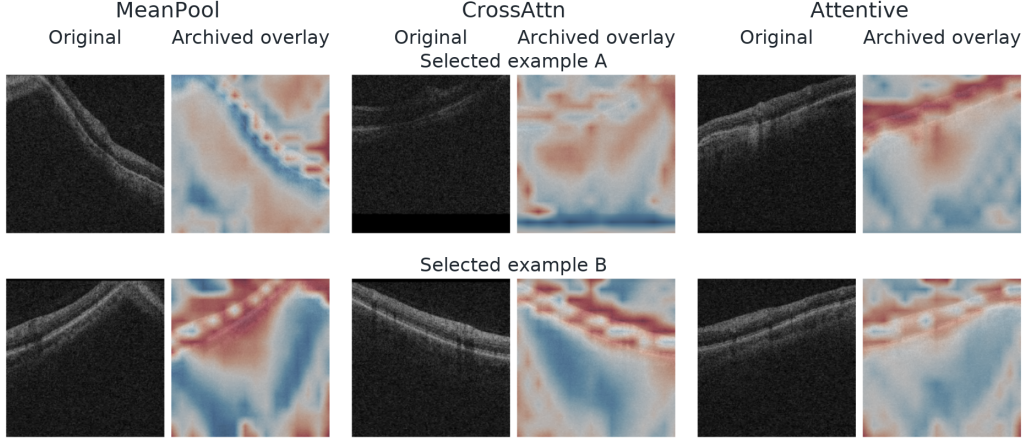
475 Positive values mean that removing the feature lowers the glaucoma logit; negative values mean
 476 that removal raises it. Both signs indicate prediction sensitivity. Figure 4 shows selected archived
 477 examples of the resulting spatial patterns.

478 The historical report and selected overlays prompted us to investigate retinal regions as prediction
 479 targets. Those examples were selected from correctly classified cases and informative slices, so
 480 they are not a population-level estimate of anatomical localization. Nor does a response to feature
 481 removal establish a clinically causal region or show which pretraining mask is best. Fine-tuning
 482 used diagnosis labels, so the maps cannot separate the contribution of self-supervised pretraining
 483 from subsequent supervised adaptation.

484 **Available evidence.** Archived occlusion curves and heat maps remain available, but the underlying
 485 per-volume numerical arrays and a complete normalization record are not retained in the evidence
 486 used for this submission. The historical record also documents sensitivity of patch-level displays
 487 to numerical precision and colour scaling. The selected image panels are retained as archived vi-
 488 sualizations, without restoring their numerical summaries as validated results. We make no spatial-
 489 consistency, laterality or clinical landmark claim from those displays. Recovering the producing
 490 arrays, checkpoint identities and plotting transforms would be necessary for a reproducible quantita-
 491 tive analysis. No new attribution campaign was run, and the saved frozen-probe AUC comparisons
 492 are unchanged.

493 E Full paired-contrast table

494 Benjamini–Hochberg is applied within a declared primary family consisting of exactly these nine
 495 contrasts (the fp16 comparisons among RANDOM, CENTROID and ENVELOPE at epochs 50, 75 and
 496 100) and the remaining 25 exploratory contrasts in the stored statistics are corrected as a separate
 497 family. The two families are never pooled, and the q values printed above are those of the nine-
 498 contrast family only. Declaring the family fixes what is corrected together. It does not make these
 499 contrasts confirmatory, because the test split was inspected repeatedly before the family was fixed,
 500 and the paper reports them as descriptive throughout.



Archived illustrations only; per-image colour scales are not comparable.
All six pairs retained. No attribution rerun, clinical validation or pixel alteration.

Figure 4: Selected archived feature-occlusion examples from fine-tuned classifiers initialized with uniform-masking I-JEPA. Each pair shows an OCT B-scan and its patch-attribution overlay. Red denotes positive $\Delta_{s,p}$ and blue negative $\Delta_{s,p}$ as defined in Equation 5. The archived plotting procedure interpolates patch values and uses a separate symmetric colour scale for each image; colour intensity is therefore not comparable across examples. Image panels are reproduced without recolouring. Original numerical headers and clinical annotations are omitted. These selected visualizations provide context for the retinal-targeting motivation, not a quantitative localization comparison.

Table 5: Primary contrasts: matched epoch *and* matched precision. Δ is a paired difference on identical test cases; the interval is a 10,000-resample paired bootstrap and p is a DeLong test for correlated ROC curves. q is Benjamini–Hochberg over these nine contrasts. All p and q values here are descriptive (Section 6). Every comparison against the null excludes zero.

contrast	epoch	Δ AUC	95% CI	p	q
ENVELOPE – RANDOM	50	+0.0120	[+0.0068, +0.0173]	<0.0001	<0.0001
ENVELOPE – RANDOM	75	+0.0080	[+0.0028, +0.0132]	0.0025	0.0045
ENVELOPE – RANDOM	100	+0.0062	[+0.0010, +0.0113]	0.0199	0.0299
CENTROID – RANDOM	50	+0.0099	[+0.0051, +0.0149]	<0.0001	0.0001
CENTROID – RANDOM	75	+0.0113	[+0.0063, +0.0164]	<0.0001	<0.0001
CENTROID – RANDOM	100	+0.0109	[+0.0057, +0.0160]	<0.0001	<0.0001
CENTROID – ENVELOPE	50	-0.0020	[-0.0069, +0.0029]	0.4114	0.4114
CENTROID – ENVELOPE	75	+0.0033	[-0.0013, +0.0079]	0.1491	0.1677
CENTROID – ENVELOPE	100	+0.0047	[+0.0004, +0.0091]	0.0294	0.0378

501 F Probe precision control

502 We repeated the recorded mixed-precision probes with autocast disabled in feature extraction, head
503 fitting and evaluation. Each comparison uses the same frozen encoder. The recorded changes are
504 smaller than the principal policy differences.

Table 6: Full fp32 re-probe against the original fp16 result, same frozen encoder in each row. The eight DeLong p -values are unadjusted diagnostics, not a family-wise test. The largest difference is 2×10^{-4} .

policy	epoch	fp16 AUC	fp32 AUC	Δ	p
ENVELOPE	50	0.876064	0.876063	-0.000001	0.964
ENVELOPE	75	0.880307	0.880305	-0.000002	0.806
ENVELOPE	100	0.880743	0.880761	+0.000018	0.167
CENTROID	50	0.874030	0.874015	-0.000015	0.412
CENTROID	100	0.885485	0.885293	-0.000192	0.767
RANDOM	50	0.864097	0.864121	+0.000024	0.128
RANDOM	75	0.872302	0.872302	-0.000001	0.962
RANDOM	100	0.874581	0.874485	-0.000096	0.000